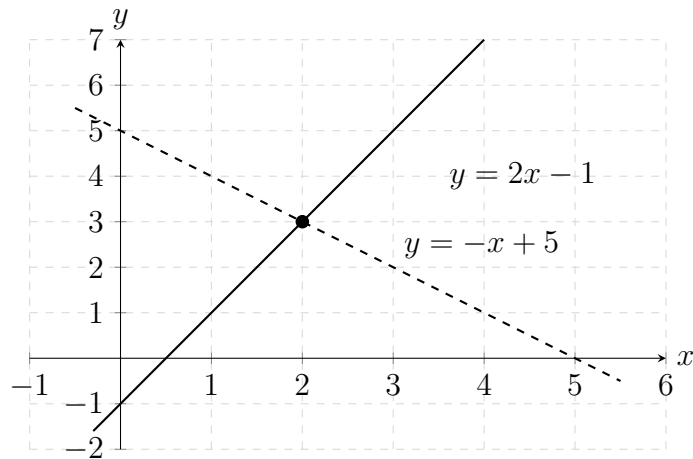


6.4 Classwork: Systems of Equations

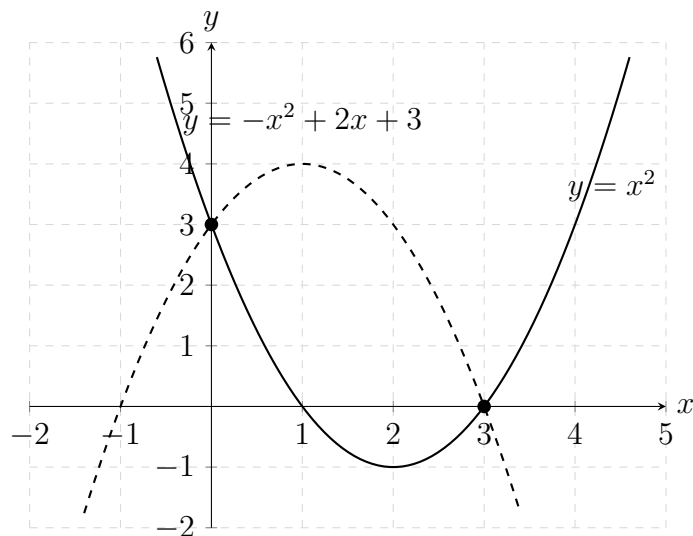
1. The diagram below shows two lines, $y = 2x - 1$ and $y = -x + 5$.



- (a) Write down the gradient of each line. [2]
- (b) Find the coordinates of the point of intersection by solving the system algebraically. [3]
- (c) Verify that the point satisfies both equations. [2]

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2. The diagram below shows two parabolas, $y = x^2 - 4x + 3$ and $y = -x^2 + 2x + 3$.



- (a) Show that the x -coordinates of the points of intersection satisfy $2x^2 - 6x = 0$. [2]
- (b) Hence find the x -coordinates of the points of intersection. [2]
- (c) Find the coordinates of both points of intersection. [2]
- (d) Find the coordinates of the vertex of $y = x^2 - 4x + 3$. [2]

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